

1) What is the main purpose of 'static block'?

- a) initialize local variables
- b) initialize static variables
- c) initialize instance variables
- d) None of above

 **Correct Answer: b) initialize static variables**

2) In java, when we implement an interface method, it must be declared as -

- a) public
- b) protected
- c) private
- d) default

 **Correct Answer: a) public**

3) What is/are the advantage(s) of "Encapsulation"?

- a) Data hiding
- b) Control over data
- c) Make a class read only and write only
- d) All of above

 **Correct Answer: d) All of above**

4) What is the purpose of "constructor" in java?

- a) Object creation
- b) Class initialization
- c) Object initialization
- d) None of above

 **Correct Answer: c) Object initialization**

5) Which type of variables are not supported by Java?

- a) Local variable
- b) Instance variable
- c) Static variable
- d) Global variable

 **Correct Answer: d) Global variable**

6) If we want to prevent a field to serialize then which modifier we should add before that field -

- a) private
- b) default
- c) final
- d) transient

 **Correct Answer: d) transient**

7) All of the variables inside "interface" is by default -

- a) public
- b) static
- c) final
- d) All of them

☒ **Correct Answer: d) All of them**

8) Which is correct one for "abstract" class in java?

- a) Can have abstract method
- b) Can have non-abstract method
- c) Can have constructor
- d) All of them

☒ **Correct Answer: d) All of them**

9) In java, a class may have ____ super class(s) and implement ____ interface(s)

- a) Only one, Only one
- b) Only one, Multiple
- c) Multiple, Only one
- d) Multiple, Multiple

☒ **Correct Answer: b) Only one, Multiple**

10) Which is the correct order to increase access privilege?

- a) public → default → protected → private
- b) private → default → protected → public
- c) default → private → protected → public
- d) private → protected → default → public

 **Correct Answer: b) private → default → protected → public**

11) What is the time complexity of Kruskal's Algorithm?

- a) $O(E + V \log V)$
- b) $O(E \log E)$
- c) $O(V^2)$
- d) $O(E^2)$

 **Correct Answer: b) $O(E \log E)$**

12) Which of the following is an in-place sorting algorithm?

- a) Merge Sort
- b) Bubble Sort
- c) Counting Sort
- d) Radix Sort

 **Correct Answer: b) Bubble Sort**

13) In a database, what is a unique identifier for each record in a table called?

- a) Secondary key
- b) Index key
- c) Primary key
- d) Foreign key

 **Correct Answer: c) Primary key**

14) What is the purpose of normalization in a database?

- a) Increase data redundancy
- b) Optimize query speed
- c) Reduce data redundancy
- d) Improve data security

 **Correct Answer: c) Reduce data redundancy**

15) How does a transaction ensure data integrity in SQL?

- a) By locking the database
- b) By allowing multiple users to modify data simultaneously
- c) By ensuring all operations within the transaction are completed before committing

d) By automatically updating all related tables

☒ **Correct Answer: c) By ensuring all operations within the transaction are completed before committing**

16) Which of the following statements is true about B+ Tree used in database indexing?

- a) Internal nodes of a B+ Tree store actual data values
- b) B+ Trees are only efficient for equality searches
- c) All data records are stored in the leaf nodes
- d) B+ Trees do not maintain sorted order of keys

☒ **Correct Answer: c) All data records are stored in the leaf nodes**

17) What is dirty read in the context of database transactions?

- a) Reading data that has been written to disk but not committed
- b) Reading data that is being read by another transaction
- c) Reading uncommitted changes made by another transaction
- d) Reading data from a corrupted database file

☒ **Correct Answer: c) Reading uncommitted changes made by another transaction**

18) What is the default capacity of an ArrayList when declared as?

- a) 5
- b) 8
- c) 10
- d) 16

 **Correct Answer: c) 10**

19) What is the worst-case time complexity to access an element in a Binary Search Tree (BST)?

- a) $O(1)$
- b) $O(\log n)$
- c) $O(n)$
- d) $O(n \log n)$

 **Correct Answer: c) $O(n)$**

20) What is the time complexity to find the diameter of a binary tree?

- a) $O(V + E)$
- b) $O(VE)$
- c) $O(V)$
- d) $O(V \log E)$

 **Correct Answer: a) $O(V + E)$**

21) Which of the following represents In order Traversal of a binary tree?

- a) Root → Left → Right
- b) Left → Root → Right
- c) Right → Root → Left
- d) Root → Right → Left

 **Correct Answer: b) Left → Root → Right**

22) What is the best time complexity to precompute all-pairs shortest paths in a weighted graph?

- a) $O(V + E)$
- b) $O(V^2)$
- c) $O(V^3)$
- d) $O(E \log V)$

 **Correct Answer: c) $O(V^3)$**

23) Which data structure is mainly used for implementing recursive algorithms?

- a) Queue
- b) Array
- c) Stack

d) Heap

☒ **Correct Answer: c) Stack**

24) How can we perform an operation without using '+' operator?

a) Using -

b) Using Bitwise Operators

c) Using Loops

d) Not possible

☒ **Correct Answer: b) Using Bitwise Operators**

25) What is the worst-case time complexity to find an object in a Hash table?

a) $O(1)$

b) $O(\log n)$

c) $O(n)$

d) $O(n \log n)$

☒ **Correct Answer: c) $O(n)$**

26) What is the key difference between HashSet and TreeSet in Java?

a) TreeSet is faster than HashSet

b) HashSet maintains sorted order

c) TreeSet maintains sorted order

d) Both are the same

 **Correct Answer: c) TreeSet maintains sorted order**